Design Fundamentals 1/4

Topics 1–6

1. What are design fundamentals?

- Four pillars: the basis (networking, storage), the key characteristics (availability, latency, throughput, redundancy, consistency), the components (LBs, caches, proxies, DBs), the tech (Kafka, Redis, S3...)

2. Client–server model

- DNS resolves to an IP
- The client opens a connection to the server's port
- The server listens
- HTTP runs on 80, HTTPS on 443

3. Network protocols

- IP packets, TCP (ordered, reliable, handshake), UDP (fast, lossy), HTTP request/response on top of TCP

4. Storage

- Memory versus disk
- Persistence
- Databases are just servers that write to disk

5. Latency and throughput

- Latency = time per operation (memory ~µs, SSD ~100 µs, cross-continent ~150 ms)
- Throughput = operations per second
- Improve them separately

6. Availability

- Nines (99.99% ≈ 52 min a year)
- SLA vs SLO
- Redundancy, no single point of failure, passive/active failover

Recap

- 1. What are design fundamentals?
- 2. Client–server model
- 3. Network protocols
- 4. Storage
- 5. Latency and throughput
- 6. Availability